
12:42
18
Sample Source
#include <iostream>
using namespace std;
int main() {
int num = 8;
for (int i = 1; i <= 10; i++) {
cout << num << " x " << i << " = "
<< num * i << endl;
}
return 0;
}
' " , . ; : = < > +
#include <iostream> u...
1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
!#1
English (UK)

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code

```
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
8 x 7 = 56
8 x 8 = 64
8 x 9 = 72
8 x 10 = 80
```

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Today

Task 1: Table of 8

Date: 15/10/2024 • Roll No: 22CS045

C++ code:

```
#include <iostream>
using namespace std;
int main() {
  int num = 8;
  for(int i=1; i<=10; i++) {
  cout << num << " x " << i << " = " << num*i << endl;
  }
  return 0;
```

* endl adds new line

Output / Answer:

```
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
8 x 7 = 56
8 x 8 = 64
8 x 9 = 72
8 x 10 = 80
```

All correct ✓ done for Lab task!

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Sample Source

```cpp
#include <iostream>
using namespace std;

int main() {
    int base, power;
    long long result = 1;

    cout << "Base enter karen: ";
    cin >> base;
    cout << "Power enter karen: ";
    cin >> power;

    for(int i = 1; i <= power; i++) {
        result = result * base;
    }

    cout << base << " ki power " << power
<< " = " << result << endl;

    return 0;
}
```

' " , . ; : = < > +

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code

```
Base enter karen: 2
Power enter karen: 3
2 ki power 3 = 8
```

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Task 2: Power of a Number (using for loop)

— Write a program to calculate power using for loop —

```
"#include <iostream>
"using namespace std;"
"int main(){
"int base, power;"
"long long result = 1;"
"cout << "Enter base: ")"
"cin >> base;"
"cout << "Enter power: ")"
"cin >> power;"
```

```
"for(int i=1; i<=power; i]}{("
"result = result * base;"
"};
"cout << base << " ^ " << power << " = " << result;"
"return 0;"
"};
```

2^3 = 8

Output example:

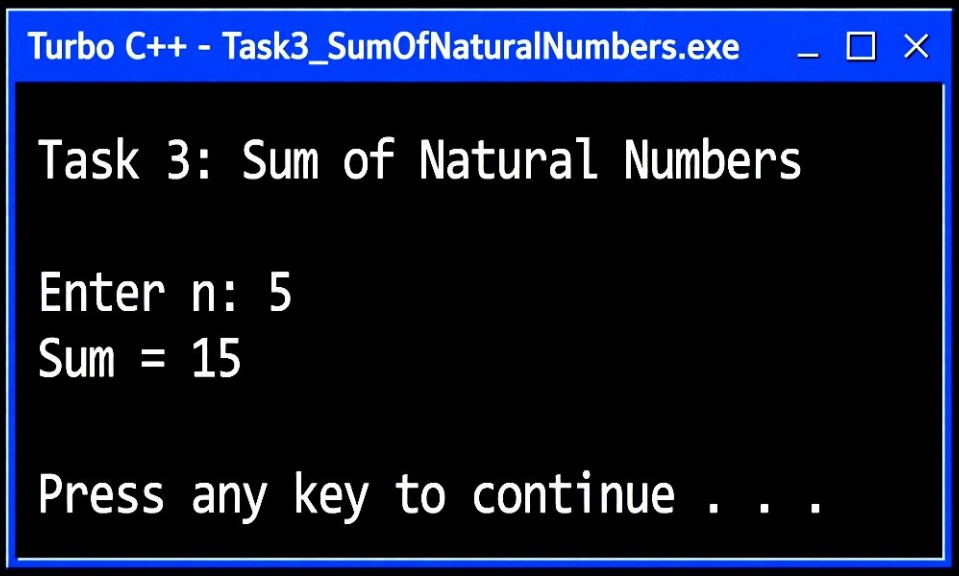
Enter base: 2
Enter power: 3
result = 8 ☆ ✓
* 2^3 = 8

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Cpp code

```cpp
#include <iostream>
using namespace std;
int main() {
    int n, sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for(int i=1; i<=n; i++){
        sum = sum + i;
    }
    cout << "Sum = " << sum;
    return 0;
}
```

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Task 3: Sum of Natural Numbers using loop

Code:

```
#include <iostream>
using namespace std;
int main() {
int n, sum=0;
cout << "Enter n: ";
cin >> n;
for(int i=1; i<=n; i]){ {
sum = sum + i;
}
cout << "Sum = " << sum;
return 0;
}
```

Output example:

Enter n: 5

* Sum = 15.

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Cpp code

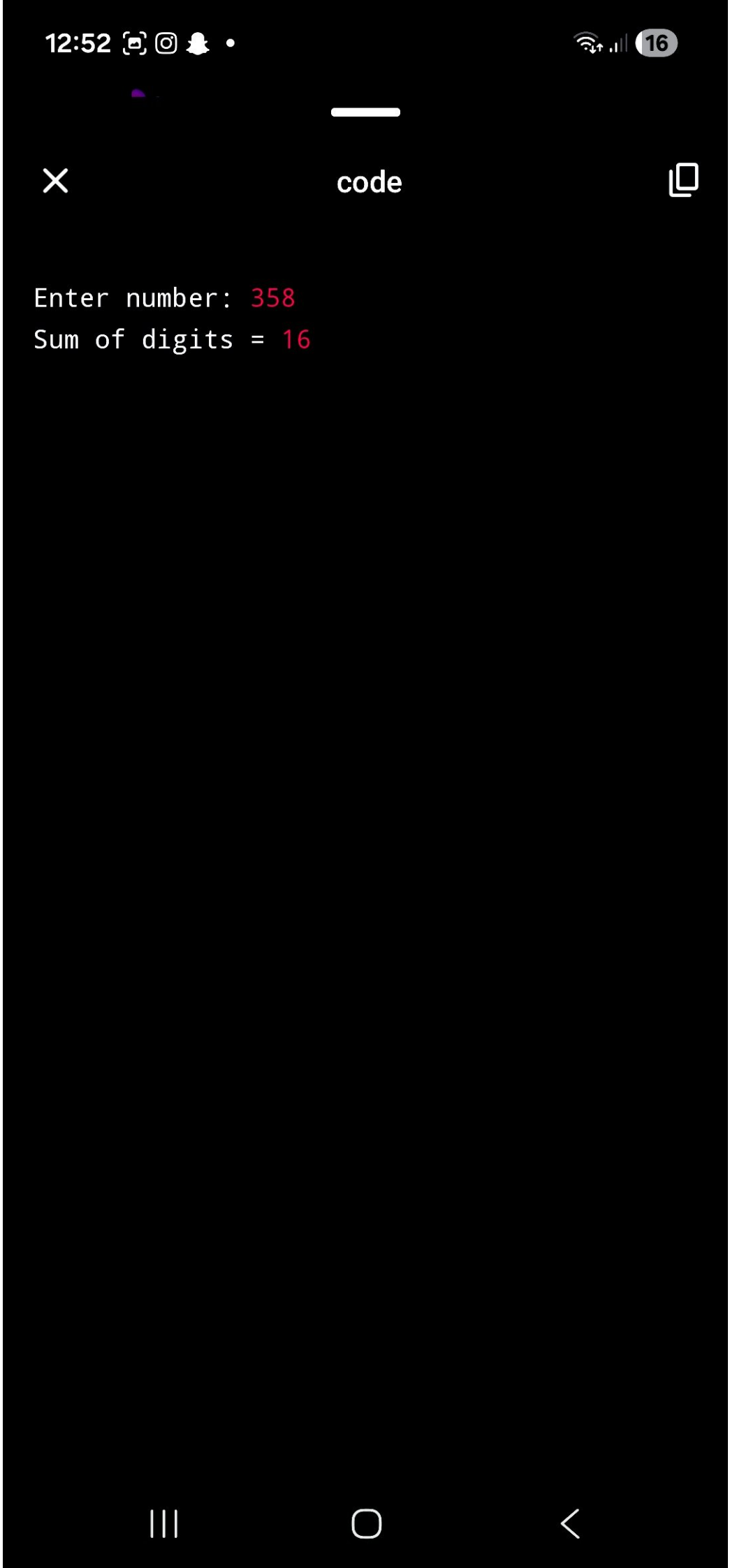
```cpp
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0, r;

    cout << "Enter number: ";
    cin >> n;

    while(n > 0) {
        r = n % 10;
        sum = sum + r;
        n = n / 10;
    }

    cout << "Sum of digits = " << sum << endl;
    return 0;
}
```


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16
code
Enter number: 358
Sum of digits = 16

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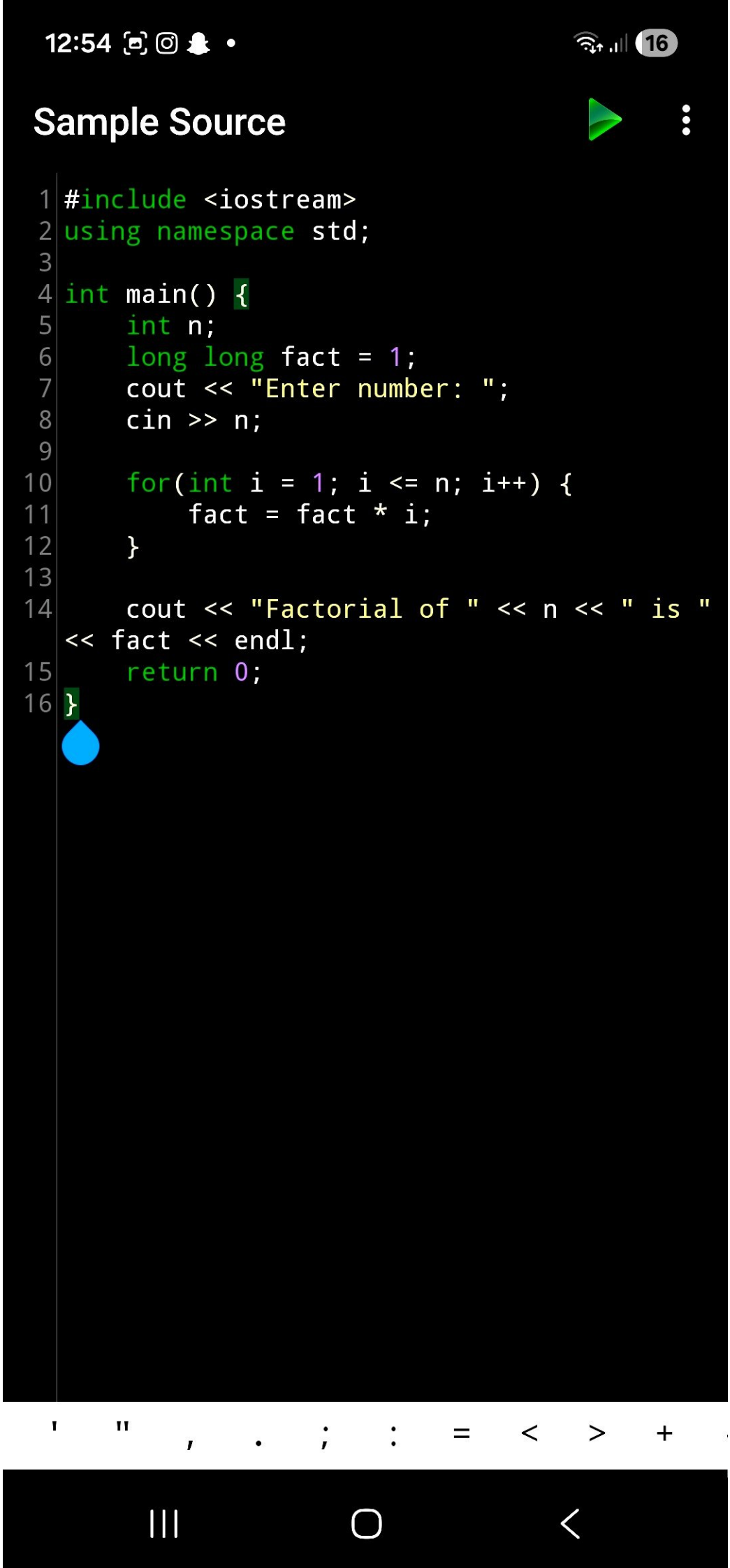
Task 4: Sum of digits

Code:

```
#include <iostream>
using namespace std;
int main() {
  int n, sum=0, r;
  cout << "Enter number: ";
  cin >> n;
  while(n>0) {
    r = n % 10;
    sum = sum + r;
    n = n / 10;
    }
    cout << "Sum of digits = " << sum;
    return 0; }
```

Output example:

Enter number: 358
Sum of digits = 16 (3+5+8=16).


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16
Sample Source
#include <iostream>
using namespace std;
int main() {
int n;
long long fact = 1;
cout << "Enter number: ";
cin >> n;
for(int i = 1; i <= n; i++) {
fact = fact * i;
}
cout << "Factorial of " << n << " is "
<< fact << endl;
return 0;
}
' " , . ; : = < > +

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code

```
Enter number: 4
Factorial of 4 is 24  (4*3*2*1=24)
```

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Task 5: Factorial of a number.

```
#include <iostream>
  using namespace std;
int main() {
- int n;
- long long fact=1;
- cout << "Enter number: ";
- cin >> n;
- for(int i=1; i<=n; i3) {
- fact=fact*i;
- }
- cout << "Factorial of " << n << " is " << fact;
} return 0;
```

Output:

Enter number: 4
Factorial of 4 is 24 (4*3*2*1=24).

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Cpp code

```cpp
#include <iostream>
using namespace std;

int main() {
    for(int i = 1; i <= 6; i++) {
        for(int j = 1; j <= i; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}
```

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code

```
*
**
***
****
*****
******
```

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Task 7: Print pattern of stars increasing.

Pattern: * - ** * *** * ***** ***** *****

Code:

```
#include <iostsreaut>
using namespose std;
int main(({
   for(int i=1:j==6:i) {
   for(int j=1:j==i:j({
   cout << "x";
   }
   cout << endl;
   }
   return 0;
   }
```

Output shows the triangle pattern.

Output: * ** *** **** ***** *****

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Cpp code

```cpp
#include <iostream>
using namespace std;

int main() {
    int month;
    cout << "Enter month number: ";
    cin >> month;

    switch(month) {
        case 1: case 3: case 5: case 7:
        case 8: case 10: case 12:
            cout << "31 days";
            break;
        case 4: case 6: case 9: case 11:
            cout << "30 days";
            break;
        case 2:
            cout << "28 or 29 days";
            break;
        default:
            cout << "Invalid month";
    }
    return 0;
}
```


Terminal — month_days.cpp
C++ Program: Month Days Checker
Enter month number: 2
28 or 29 days.
Enter month number: 12
31 days.
Enter month number: _

12:57 15
Battery power 15% Charge your phone or tap h...

Task: Month Days Checker using switch case

```
#include <iostream>
using nemepose std:
int main> : int month>.
cout <= "Enter month number : ≠
cin => month>.
swie 1:(mogth>
case 3:
case 5:
case 8:
case 10:
case 12: "31 days"
case 4:      6:
case 11: "30 days"
case 2: "28 or 29 days"
dafdet: "Invatid month"
retver t 0.
```

Output:

Enter month number: 2 -> 28 or 29 days.
Enter month number: 12 -> 31 days.

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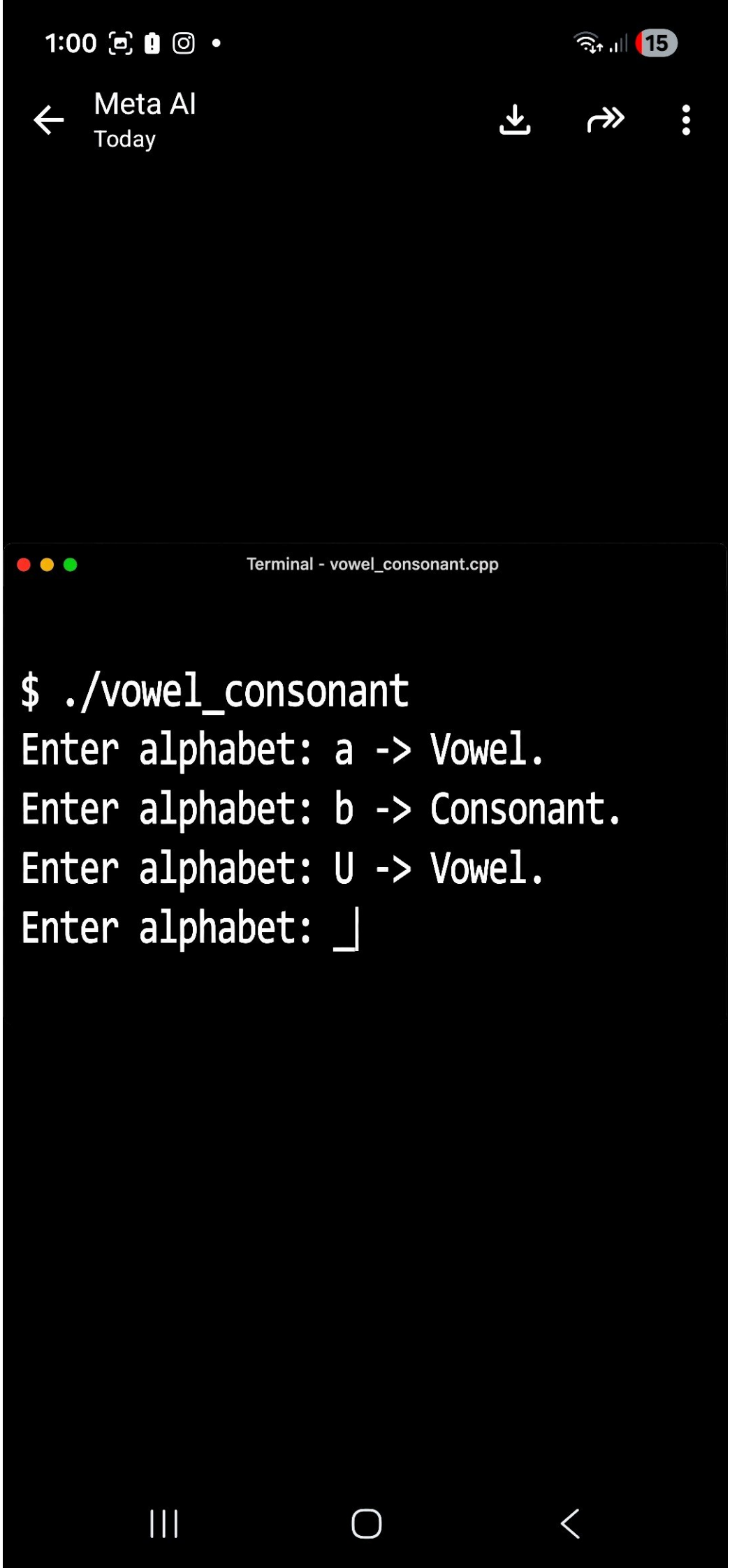
Sample Source

```cpp
#include <iostream>
using namespace std;

int main() {
    char ch;
    cout << "Enter alphabet: ";
    cin >> ch;

    switch(ch) {
        case 'a': case 'e': case 'i': case 'o': case 'u':
        case 'A': case 'E': case 'I': case 'O': case 'U':
            cout << "Vowel";
            break;
        default:
            cout << "Consonant";
    }
    return 0;
}
```

' " , . ; : = < > +


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Terminal - vowel_consonant.cpp
$ ./vowel_consonant
Enter alphabet: a -> Vowel.
Enter alphabet: b -> Consonant.
Enter alphabet: U -> Vowel.
Enter alphabet:

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C++ Program -> Check if Alphabet is Vowel or Consonant (using switch case)

```
#include <iostream>
 using namepsace std;
 int main(h{
  - char ch;
     cout << "Enter alphabet: '; cin >> ch;
  - switch(ch) {
      case 'a';
      case 'e';
      case 'i';
      case 'o';
      case 'u';
      case 'A';
      case 'E';
      case 'I';
      case 'O';
      case 'U';
    cout << "Vowel"; break;
    default:
     - cout << "Consonant";
       }
     return 0;
     } // end of main
```

Output:

Enter alphabet: a -> Vowel
Enter alphabet: b -> Consonant.